



AQ4.2: Activity Questions 2 - Not Graded



This assignment will not be graded and is only for practice.

Level 1

1 point

Match the vector spaces (with the usual scalar multiplication and vector addition as in $M_{3 \times 3}(R)$) in column A with their bases in column B in Table : M2W4AQ2.

Choose the correct option.

☐

a $\rightarrow$ $\rightarrow$ ii

☐

a $\rightarrow$ $\rightarrow$ i

☐

b $\rightarrow$ $\rightarrow$ iii

☐

c $\rightarrow$ $\rightarrow$ i

☐

b $\rightarrow$ $\rightarrow$ ii

☐

c $\rightarrow$ $\rightarrow$ iii

No, the answer is incorrect.

Score: 0

Accepted Answers:

a $\rightarrow$ $\rightarrow$ ii

b $\rightarrow$ $\rightarrow$ iii

c $\rightarrow$ $\rightarrow$ i

If S be a subset of R^5 such that $\text{span}(S) = R^5$, then what is the minimum number of elements possible in S ?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 5

1 point

1 point

If S_1 is a maximal linear independent set and S_2 is a minimal spanning set of a vector space V , then which of the following option(s) is (are) true?

☐

For any $v \in S_1$, $S_1 \setminus \{v\}$ is a linearly independent.

☐

For any $v \in S_2$, $S_2 \setminus \{v\}$ is a spanning set of V .

☐

For any $v \in V \setminus S_1$, $S_1 \cup \{v\}$ is a linearly dependent.

☐

For any $v \in V$, $S_2 \cup \{v\}$ is a spanning set of V .

No, the answer is incorrect.

Score: 0

Accepted Answers:



For any $v \in S_1$, $v \in S_1$, $S_1 \setminus \{v\}$ is a linearly independent.

For any $v \in V \setminus S_1$, $v \in V \setminus S_1$, $S_1 \cup \{v\}$ is a linearly dependent.

For any $v \in V$, $S_2 \cup \{v\}$ is a spanning set of V .

1 point

Consider the following subset of R^2 with usual addition and scalar multiplication as in R^2 .

- $V_1 = \{(x, 0) \mid x \in R\}$
- $V_2 = \{(0, y) \mid y \in R\}$

Which of the following options are correct?

☐

Both V_1 and V_2 are subspaces of R^2 .

☐

$V_1 \cap V_2$ is a subspace of R^2 .

☐

$V_1 \cup V_2$ is a subspace of R^2 .

☐

V_1 is a subspace of R^2 , but V_2 is not.

☐

V_2 is a subspace of R^2 , but V_1 is not.

☐

$\{(1, 0)\}$ is a basis of V_1 .

☐

$\{(0, 1)\}$ is a basis of V_2 .

No, the answer is incorrect.

Score: 0

Accepted Answers:

Both V_1 and V_2 are subspaces of R^2 .

$V_1 \cap V_2$ is a subspace of R^2 .

$\{(1, 0)\}$ is a basis of V_1 .

$\{(0, 1)\}$ is a basis of V_2 .

1 point

Consider the following subset of R^2 with usual addition and scalar multiplication as in R^2 .

- $V_1 = \{(x, 0) \mid x \in R\}$
- $V_2 = \{(2x, 0) \mid x \in R\}$

Which of the following options are correct?

☐

Both V_1 and V_2 are subspaces of R^2 .

☐

$V_1 \cap V_2$ is a subspace of R^2 .

☐

$V_1 \cup V_2$ is a subspace of R^2 .

☐

V_1 is a subspace of R^2 , but V_2 is not.

☐

V_2 is a subspace of R^2 , but V_1 is not.

☐

$\{(1, 0)\}$ is a basis of V_1 .

☐

$\{(1, 0)\}$ is a basis of V_2 .

No, the answer is incorrect.

Score: 0

Accepted Answers:

Both V_1 and V_2 are subspaces of $\mathbb{R}^2$.

$V_1 \cap V_2$ is a subspace of $\mathbb{R}^2$.

$V_1 \cup V_2$ is a subspace of $\mathbb{R}^2$.

$\{(1, 0)\}$ is a basis of V_1 .

$\{(1, 0)\}$ is a basis of V_2 .

1 point

Let V be a vector space which is defined as follows:

$$V = \{(x, y, z, w) \mid x + z = y + w\} \subseteq \mathbb{R}^4$$

with usual addition and scalar multiplication. Which of the following set forms a basis of V ?



$\{(1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 0, 1)\}$



$\{(1, 1, 0, 0), (0, 1, -1, 0), (0, -1, 0, 1)\}$



$\{(1, 0, -1, 0), (1, 1, 0, 0), (1, 0, 0, 1)\}$



$\{(1, 1, 0, 0), (0, 1, 1, 0), (0, -1, 0, 1)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(1, 0, -1, 0), (1, 1, 0, 0), (1, 0, 0, 1)\}$

$\{(1, 1, 0, 0), (0, 1, 1, 0), (0, -1, 0, 1)\}$

Level 2

1 point

Which of the following sets form a basis of the vector space of 2×2 lower triangular real matrices with usual matrix addition and scalar multiplication? (More than one option may be correct)



$\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \right\}$



$\left\{ \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right\}$



$\left\{ \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \right\}$



$\left\{ \begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \right\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \right\}$

$\left\{ \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right\}$

$\left\{ \begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \right\}$

If SS is a linearly independent set in vector space $V = \{(x, y) \mid y = 2x, \text{ where } x, y \in \mathbb{R}\}$ with usual addition and scalar multiplication as in $\mathbb{R}^2$, then find the maximum possible cardinality of SS .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

1 point

If $\{v_1, v_2, v_3\}$ forms a basis of $\mathbb{R}^3$, then which of the following are true?

☐

$\{v_1, v_2, v_1 + v_3\}$ forms a basis of $\mathbb{R}^3$

☐

$\{v_1, v_1 + v_2, v_1 + v_3\}$ forms a basis of $\mathbb{R}^3$.

☐

$\{v_1, v_1 + v_2, v_1 - v_3\}$ forms a basis of $\mathbb{R}^3$.

☐

$\{v_1, v_1 - v_2, v_1 - v_3\}$ forms a basis of $\mathbb{R}^3$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{v_1, v_2, v_1 + v_3\}$ forms a basis of $\mathbb{R}^3$

$\{v_1, v_1 + v_2, v_1 + v_3\}$ forms a basis of $\mathbb{R}^3$.

$\{v_1, v_1 + v_2, v_1 - v_3\}$ forms a basis of $\mathbb{R}^3$.

$\{v_1, v_1 - v_2, v_1 - v_3\}$ forms a basis of $\mathbb{R}^3$.

1 point

Which of the following options is(are) true?

☐

Any minimal spanning set of a vector space V must be a basis of V .

☐

Any maximal spanning set of a vector space V must be a basis of V .

☐

Any minimal linear independent set of vector space V must be a basis of V .

☐

Any maximal linear independent set of vector space V must be a basis of V .

☐

The basis of a vector space is unique.

☐

The number of elements in a basis of $\mathbb{R}^3$ is 33.

☐

The number of elements in a basis of $M_{3 \times 3}(\mathbb{R})$ is 33.

☐

There are infinite number of bases of $\mathbb{R}^3$.

☐

Any subset of a minimal spanning set of V cannot be a spanning set.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Any minimal spanning set of a vector space V must be a basis of V .

Any maximal linear independent set of vector space V must be a basis of V .

The number of elements in a basis of $\mathbb{R}^3$ is 33.

There are infinite number of bases of R^3 .

Any subset of a minimal spanning set of V cannot be a spanning set.

Check Answers

Your score is: 0/10